G12Adv	Mathematics	الرياضيات	T1 - 2017/18

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

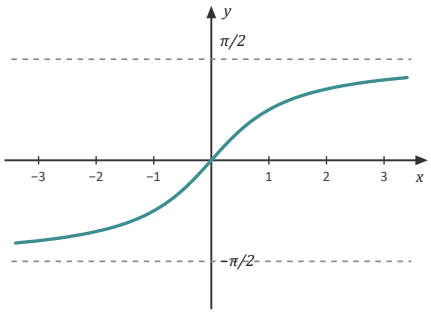
1. Find $\lim_{x\to 0}\left(\frac{x^2}{\cos^2 x - 1}\right)$ if it exists.

A) -1 **B)** 0

C) 1 **D)** ∞

2. Evaluate

$$\lim_{x\to\infty} \sin(\tan^{-1} x)$$



A) $\frac{\pi}{2}$ **B)** $-\frac{\pi}{2}$

C) -1 **D)** 1

3. If $f(x) = 2x - x^5 + 1$, find $f''(-1)$.

A) $f''(-1) = -20$ **B)** $f''(-1) = 0$

C) $f''(-1) = 20$ **D)** $f''(-1) = -3$

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4. **Evaluate**

$$\lim_{x\to\infty} \frac{6x^2-6}{3x^3+2x+1}$$

A) 0 **B)** 2

C) 3 **D)** ∞

5. **Find the average velocity between $t=0$ and $t=1$ for the position function $s(t)=\sqrt{t^2+8t}$, where s in meters and t in seconds.**

A) $\frac{5}{3}$ m/s

B) 3 m/s

C) -3 m/s

D) 0 m/s

6. **Determine the interval(s) where $f(x)=\ln(3x-6)$ is continuous.**

A) $(-2,\infty)$ **B)** $[2,\infty)$

C) $(-\infty,2)$ **D)** $(2,\infty)$

7. **Find any slant asymptotes of $f(x)=\frac{x^2+1}{x-2}$.**

A) $y=-2$ **B)** $y=2$

C) $y=x+2$ **D)** $y=x-2$

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8. If $f(x) = \dfrac{3}{2x+1}$ find $f'(x)$.

A) $f'(x) = \dfrac{3}{(2x+1)^2}$

B) $f'(x) = \dfrac{6}{(2x+1)^2}$

C) $f'(x) = \dfrac{-3}{(2x+1)^2}$

D) $f'(x) = \dfrac{-6}{(2x+1)^2}$

9. Find the derivative of the function $f(x) = e^x \ln x$.

A) $f'(x) = xe^x$

B) $f'(x) = \dfrac{e^x}{x} + e^x \ln x$

C) $f'(x) = \dfrac{e^x}{x} + \ln x$

D) $f'(x) = e^x + \dfrac{1}{x}$

10. Determine the function which is differentiable at $x = 2$.

A) $f(x) = \begin{cases} 4x, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$

B) $g(x) = \begin{cases} 4, & x < 2 \\ 2x, & x \geq 2 \end{cases}$

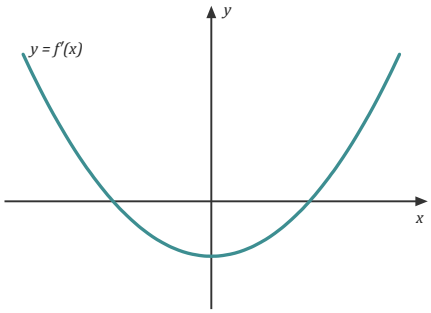
C) $p(x) = \begin{cases} 4 + 2x, & x < 2 \\ 2x, & x \geq 2 \end{cases}$

D) $h(x) = \begin{cases} 3x, & x < 2 \\ x + 4, & x \geq 2 \end{cases}$

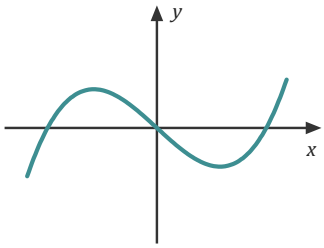
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Multiple Choice	اختيار من متعدد

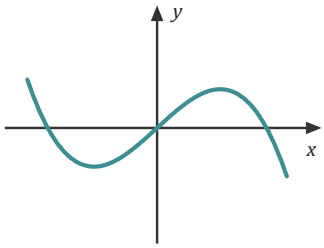
11. **Use the given graph of $f'(x)$ to determine a plausible graph of the continuous function $f(x)$.**



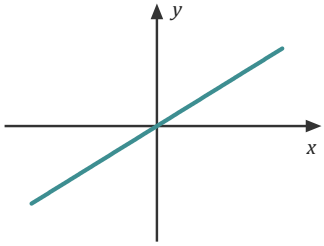
A)



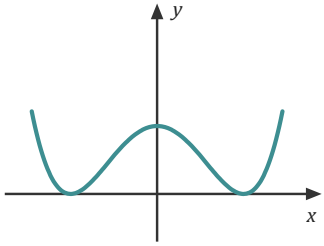
B)



C)



D)



12. **Evaluate**

$$\lim_{x\to 1}\left(\frac{x^2-1}{\sqrt{10-x}-3}\right)$$

A) -12

B) -6

C) 6

D) 12

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13. Determine the values of a and b that make the given function continuous.

$$f(x)=\begin{cases} a\,e^{x}+1, & x<0 \\ \sin^{-1}\dfrac{x}{2}, & 0\le x\le 2 \\ x^{2}-x+b, & x>2 \end{cases}$$

A) $a=-1,\ b=\dfrac{\pi}{2}-2$

B) $a=1,\ b=\dfrac{\pi}{2}-2$

C) $a=-1,\ b=\dfrac{\pi}{2}+2$

D) $a=1,\ b=\dfrac{\pi}{2}+2$

14. Use the definition of the limit to find the derivative of $f(x)=x^{2}-2x$ at $x=3$.

A) -4

B) 4

C) 3

D) -3

15. Given that the function $f(x)=x^{3}+2x+1$ has an inverse function g, compute $g'(-2)$.

A) $g'(-2)=-1$

B) $g'(-2)=5$

C) $g'(-2)=\frac{1}{5}$

D) $g'(-2)=0$

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16. **Find the derivative of the function** $f(x) = \sqrt{\tan(x^3 + 2x)}$.

A) $f'(x) = \dfrac{(3x^2 + 2)\sec^2(x^3 + 2x)}{2\sqrt{\tan(x^3 + 2x)}}$

B) $f'(x) = \dfrac{(3x^2 + 2)\sec^2(x^3 + 2x)}{\sqrt{\tan(x^3 + 2x)}}$

C) $f'(x) = \dfrac{(x^3 + 2)\sec^2(x^3 + 2x)}{2\sqrt{\tan(x^3 + 2x)}}$

D) $f'(x) = \dfrac{(3x^2 + 2)\sec^2(x^3 + 2x)}{3\sqrt{\tan(x^3 + 2x)}}$

17. **Find all points for which the tangent line to** $x^2y^2 = 3y + 1$ **is horizontal.**

A) $\left(0, -\dfrac{1}{3}\right)$

B) $\left(0, \dfrac{1}{3}\right)$

C) $\left(-\dfrac{1}{3}, 0\right)$

D) $\left(\dfrac{1}{3}, 0\right)$

18. **Suppose the price of an object is AED 12 and 10,000 units are sold. The company wants to increase the quantity sold by 1000 units per year, while increasing the revenue by AED 15,000 per year. At what rate would the price have to be increased to reach these goals?**

A) AED 3 per year

B) AED 5 per year

C) AED 0.5 per year

D) AED 0.3 per year

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19. **Find a value of c satisfying the conclusion of the Mean Value Theorem for $f(x) = x^2 + 2x + 1$ on the interval $[0, 1]$.**

A) $c = 1$

B) $c = \frac{1}{2}$

C) $c = 0$

D) $c = \frac{1}{3}$

20. **Find the linear approximation for $f(x) = \sqrt{2x + 9}$ at $x_0 = 0$.**

A) $L(x) = \frac{2}{3}x + 3$

B) $L(x) = \frac{1}{6}x + 3$

C) $L(x) = \frac{1}{3}x + 3$

D) $L(x) = \frac{1}{3}x + 2$